MODULE 11 · DOMAIN 2 - ACCOUNT MANAGEMENT AND DATA GOVERNANCE

Credits, resource monitors and the usage views

What things cost, how to cap them, and where to look afterwards.

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Warehouse credits per hour

SIZE	CREDITS/HOUR
X-Small	1
Small	2
Medium	4
Large	8
X-Large	16
2X-Large through 6X-Large	32, 64, 128, 256, 512

ON THE EXAM

Each size doubles the one below. Billed per second, with a 60-second minimum on every start or resume.

What actually consumes credits

- **Virtual warehouse compute.** The table above. The largest line on most bills.
- **Serverless compute.** Snowpipe, automatic clustering, materialized view maintenance, serverless tasks, search optimization.
- **Cloud services**, above the 10% daily allowance. Storage is separate and billed in currency, not credits.
- **Serverless is the category people forget**, and it is the one resource monitors cannot control.

> ON THE EXAM
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> Every cost question starts with working out which of these meters the described activity lands on.

Resource monitors

- **Only ACCOUNTADMIN can create or own one.** That is a hard requirement, not a recommendation.
- **One account-level monitor per account.** You may have many warehouse-level monitors.
- Configured with a **CREDIT_QUOTA** and a **FREQUENCY**: monthly by default, or daily, weekly, yearly, or never.
- Usage **resets at midnight UTC**, whatever time you gave in START_TIMESTAMP.
- If an **END_TIMESTAMP** is reached, assigned warehouses are suspended **even if the quota was never hit**.

> ON THE EXAM
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> A monitor with no CREDIT_QUOTA never reaches 100%, so its triggers never fire. An easy configuration to get wrong.

The three trigger actions

ACTION	WHAT HAPPENS
NOTIFY	A notification is sent. No action is taken on any warehouse.
SUSPEND	Notify, then suspend assigned warehouses after running statements have finished.
SUSPEND_IMMEDIATE	Notify and suspend at once, cancelling any statements currently executing.

ON THE EXAM

Thresholds above 100% are allowed. A common pattern is notify at 75, suspend at 100, suspend immediately at 110.

What resource monitors cannot do

- **They cannot track serverless features or AI services at all.** Use a **budget** for those.
- An **account-level** monitor does not control serverless credit usage. Snowpipe and automatic reclustering run on regardless.
- A **warehouse-level** monitor can **monitor but not suspend** cloud services usage.
- The quota **ignores the 10% cloud services adjustment**, counting credits you are never billed for.
- Suspension takes time even when immediate. Snowflake advises **buffers** and **one warehouse per monitor** for strict control.

> ON THE EXAM
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> This slide is the module's highest-value content. Most resource monitor questions are really about a limitation.

Getting a suspended warehouse back

- Once a suspend trigger fires, assigned warehouses **cannot be resumed** until one of five things happens.
- **The next interval starts**, as dictated by the monitor's frequency and start date.
- **The credit quota is increased**, or **the suspend threshold is increased**.
- **The warehouse is unassigned** from the monitor, or **the monitor is dropped**.
- Simply issuing ALTER WAREHOUSE … RESUME will not work while the trigger is still active.

> ON THE EXAM
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> That is a real operational scenario: a warehouse that will not start, and no error explaining why.

ACCOUNT_USAGE or INFORMATION_SCHEMA?

ACCOUNT_USAGE

In the shared SNOWFLAKE database

- Includes dropped objects
- Latency of 45 minutes to 3 hours
- Retention of 1 year, 365 days
- Account-wide, ACCOUNTADMIN by default

INFORMATION_SCHEMA

In every database

- Dropped objects are not included
- No latency at all
- Retention of 7 days to 6 months
- Scoped to its database, plus some account functions

THE TIEBREAKER

Two questions settle it. How far back do you need to look, and do you need the last few minutes?

The same name, two very different answers

VIEW OR FUNCTION	WHERE, AND HOW FAR BACK
QUERY_HISTORY	365 days as an Account Usage view. 7 days as an Information Schema table function.
WAREHOUSE_METERING_HISTORY	Credits consumed per warehouse, per hour. Account Usage.
AUTOMATIC_CLUSTERING_HISTORY	Serverless credits spent reclustering, per table.
METERING_HISTORY	Credits by service type. Where serverless and classification costs appear.
ACCESS_HISTORY	Who read and wrote which columns. Enterprise Edition.
COPY_HISTORY	Data loading history. Covered again in module 14.

ON THE EXAM

QUERY_HISTORY is the classic trap. Same name, same columns, and 365 days against 7.

Budgets, and the wider organization

- A **budget** monitors credit consumption including **serverless and AI services**, which resource monitors cannot see.
- **ORGANIZATION_USAGE** covers every account in the organization. Latency is typically **24 hours**.
- Reached from the organization account, or a regular account with **ORGADMIN** enabled.
- **READER_ACCOUNT_USAGE** covers reader accounts you have created, with an extra READER_ACCOUNT_NAME column. Empty if you have none.

> ON THE EXAM
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> Three scopes: one warehouse, one account, one organization. Match the scope to what the question is asking about.

How this module gets tested

THE QUESTION SHAPE	THE ANSWER
'Cap this warehouse's spending'	Resource monitor with a quota and triggers.
'Snowpipe costs rose and the monitor did nothing'	Monitors cannot see serverless. Use a budget.
'The warehouse will not resume'	A suspend trigger is active. Raise the quota or threshold.
'Audit activity from nine months ago'	ACCOUNT_USAGE. INFORMATION_SCHEMA is too short.
'What happened in the last ten minutes'	INFORMATION_SCHEMA. ACCOUNT_USAGE lags by hours.
'Find the table that was dropped'	ACCOUNT_USAGE, which retains dropped objects.

ON THE EXAM

The last three are one distinction asked three ways: retention, latency, dropped objects.

An account-level resource monitor has a quota of 10,000 credits with a suspend trigger at 100 percent. Finance reports that spending exceeded the quota by a noticeable margin last month, and no warehouse was suspended.

Investigation shows the overspend came from Snowpipe ingestion and automatic clustering on several large tables.

Why did the monitor not act, and what should be used instead?

A The trigger threshold was too high. Set it to 90 percent so it fires earlier.

B Account-level monitors only cover the default warehouse. Add a warehouse-level monitor to each.

C The monitor was not owned by ACCOUNTADMIN, so its triggers were ignored.

D Resource monitors cannot track serverless features. A budget is required for that spending.

◆ PAUSE AND DECIDE

Why did the monitor not act, and what should be used instead?

A Lower the threshold to 90 percent.

A percentage of a quota that never counted this spending. The trigger cannot fire on credits the monitor cannot see.

B Account monitors cover only the default warehouse.

Invented. An account-level monitor covers all warehouses. It is serverless compute it cannot see, not other warehouses.

C Not owned by ACCOUNTADMIN.

Only ACCOUNTADMIN can create one, so this monitor necessarily was. Ownership does not silently disable triggers.

D Monitors cannot track serverless. Use a budget.

Snowpipe and automatic clustering are serverless. The documentation directs you to budgets for exactly this.

THE SIGNAL

'Snowpipe and automatic clustering' is the whole question. Both are serverless, and both are invisible to a resource monitor.

A reporting warehouse suspended itself mid-morning and will not restart. An engineer runs ALTER WAREHOUSE reporting_wh RESUME repeatedly, which appears to succeed, but the warehouse is suspended again moments later.

The warehouse is assigned to a monthly resource monitor whose suspend trigger fired at 100 percent of its quota. It is the tenth of the month.

What will actually restore service?

A Nothing until the first of next month, when the monthly interval resets.

B Increase the credit quota, or raise the suspend threshold, or unassign the warehouse from the monitor.

C Change the trigger from SUSPEND to SUSPEND_IMMEDIATE, which releases the current suspension.

D Resize the warehouse, since a resized warehouse is treated as a new warehouse by the monitor.

◆ PAUSE AND DECIDE

What will actually restore service?

A Nothing until next month.

The next interval is one of the five remedies, and it is not the only one. Twenty days of no reporting is not the answer.

B Raise the quota, raise the threshold, or unassign.

Three of the five documented ways to release the suspension. Dropping the monitor is a fourth.

C Switch to SUSPEND_IMMEDIATE.

Changes how future suspensions behave, and makes them harsher. It does not release the current one.

D Resize the warehouse.

Invented. Resizing does not create a new object, and the assignment to the monitor persists.

THE SIGNAL

The repeated RESUME appearing to succeed is realistic. The trigger is still active, so it suspends again immediately.

> A compliance investigation needs a list of every query run against a table over the previous nine months, including queries against two tables that have since been dropped.
>
> An engineer proposes using the QUERY_HISTORY table function in the INFORMATION_SCHEMA of the relevant database.

What is wrong with the proposal?

A. INFORMATION_SCHEMA retains 7 days and excludes dropped objects. Use the ACCOUNT_USAGE view instead.

B. Nothing is wrong. QUERY_HISTORY behaves identically in both schemas.

C. QUERY_HISTORY does not exist in INFORMATION_SCHEMA. It is an ACCOUNT_USAGE view only.

D. ACCOUNT_USAGE would also fail, since its retention is 90 days. The data is unrecoverable.

◆ PAUSE AND DECIDE

What is wrong with the proposal?

A 7 days, and no dropped objects. Use ACCOUNT_USAGE.

Both requirements fail in INFORMATION_SCHEMA. ACCOUNT_USAGE gives 365 days and retains records for dropped objects.

B They behave identically.

Identical names and columns, and very different retention. That similarity is exactly what makes the trap work.

C It does not exist in INFORMATION_SCHEMA.

It does, as a table function. The problem is its retention window, not its existence.

D ACCOUNT_USAGE retains only 90 days.

ACCOUNT_USAGE retains 1 year. Ninety days is the maximum Time Travel figure, which is a different feature entirely.

THE SIGNAL

Two independent failures: nine months exceeds 7 days, and dropped objects are excluded. Either alone rules it out.

A nightly transformation takes four hours on a Large warehouse. A team proposes moving it to X-Large to finish sooner, and a manager objects that this will double the cost.

The job is a single long-running query over a very large table, with no queueing and no spilling to remote storage.

How should the cost be assessed?

A The manager is right. X-Large always costs twice as much for the same work.

B Cost falls automatically, because larger warehouses receive a volume discount per credit.

C If runtime halves, credits are roughly unchanged: 16 per hour for 2 hours equals 8 per hour for 4.

D Cost is irrelevant, because the 60-second minimum dominates billing for long-running jobs.

◆ PAUSE AND DECIDE

How should the cost be assessed?

A X-Large always costs twice as much.

True per hour, and the hours are what change. It assumes runtime stays the same, which is the thing being disputed.

B A volume discount per credit.

Invented. A credit costs the same whatever size consumed it.

C If runtime halves, credits are roughly unchanged.

16 x 2 equals 8 x 4. Doubling size doubles the rate and, for a well-parallelised job, roughly halves the time.

D The 60-second minimum dominates.

The minimum matters for very short queries. Over four hours it is negligible.

THE SIGNAL

'No queueing and no spilling' says the job is not constrained in a way resizing fixes. Scaling is rarely perfectly linear.

What to carry into the next module

- **Each warehouse size doubles the credits of the one below.** XS is 1, 6XL is 512.
- **Resource monitors cannot see serverless or AI spending.** That is what budgets are for.
- **SUSPEND waits for running statements. SUSPEND_IMMEDIATE cancels them.** Thresholds above 100% are allowed.
- **ACCOUNT_USAGE:** dropped objects, 45 min to 3 hours latency, 365 days.
  INFORMATION_SCHEMA: none, none, 7 days to 6 months.
- **Only ACCOUNTADMIN** creates resource monitors. One account-level monitor per account.

> ON THE EXAM
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> Next: the Domain 2 drill. Six worked scenarios across modules 8 to 11, and no new content.